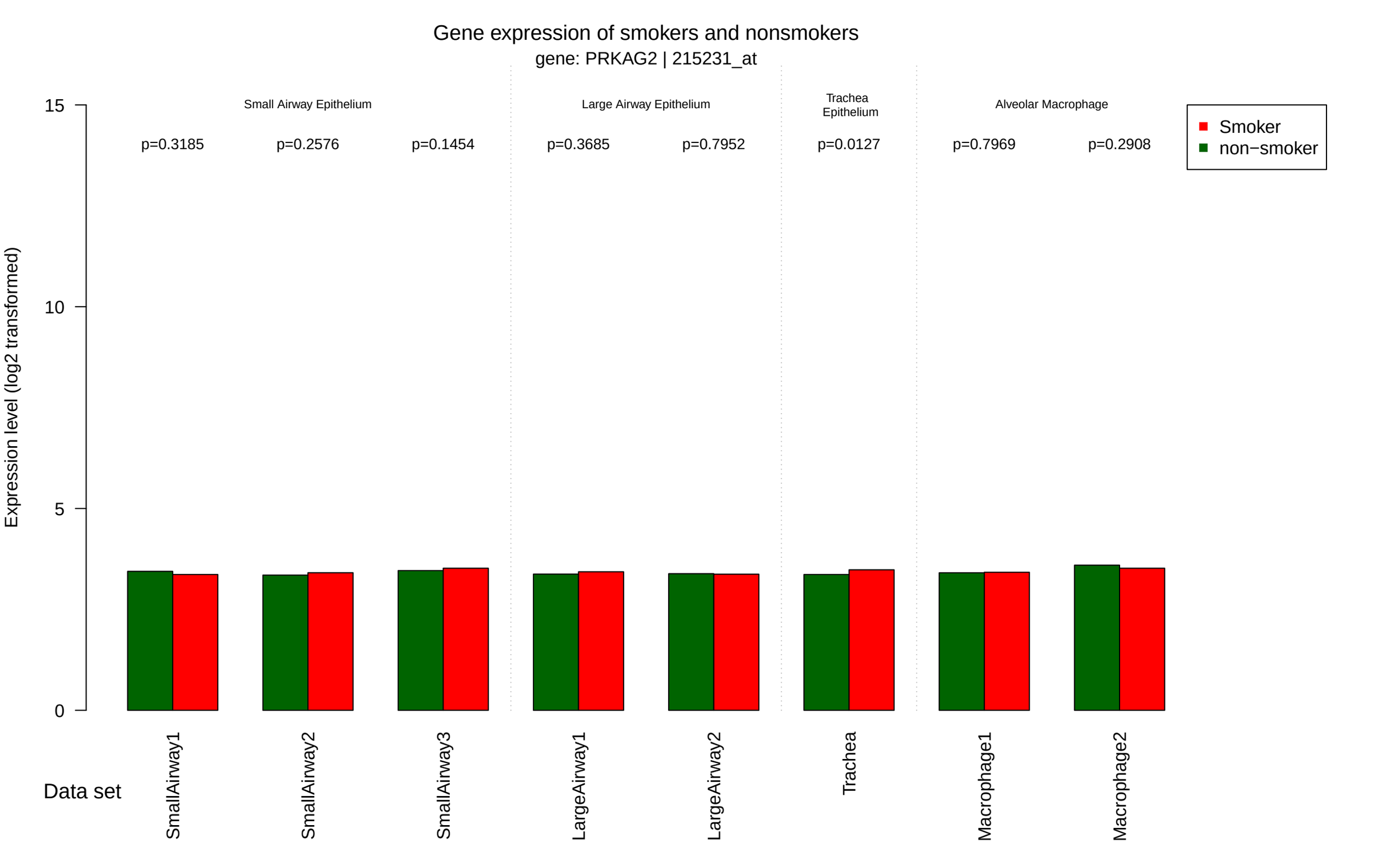
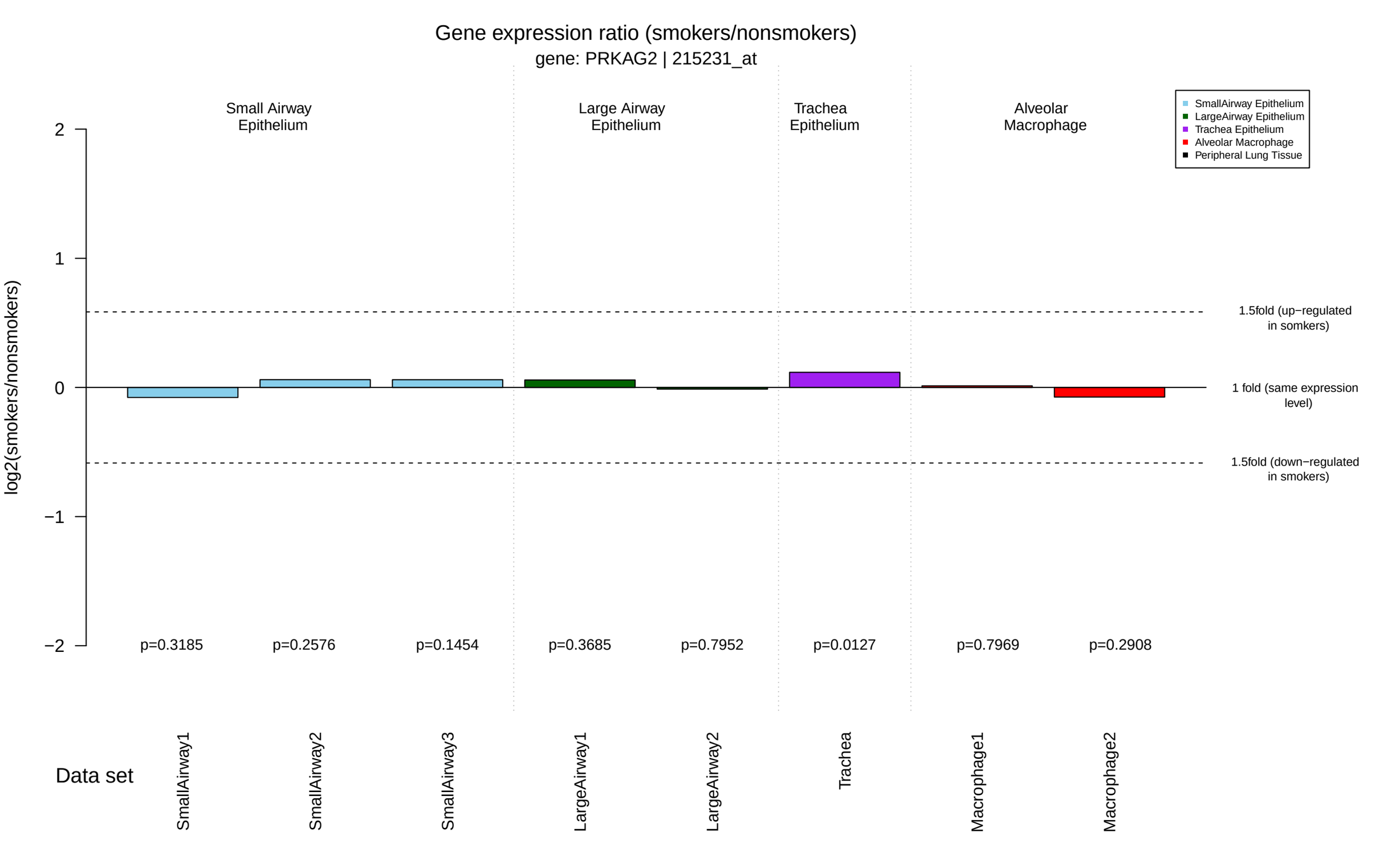
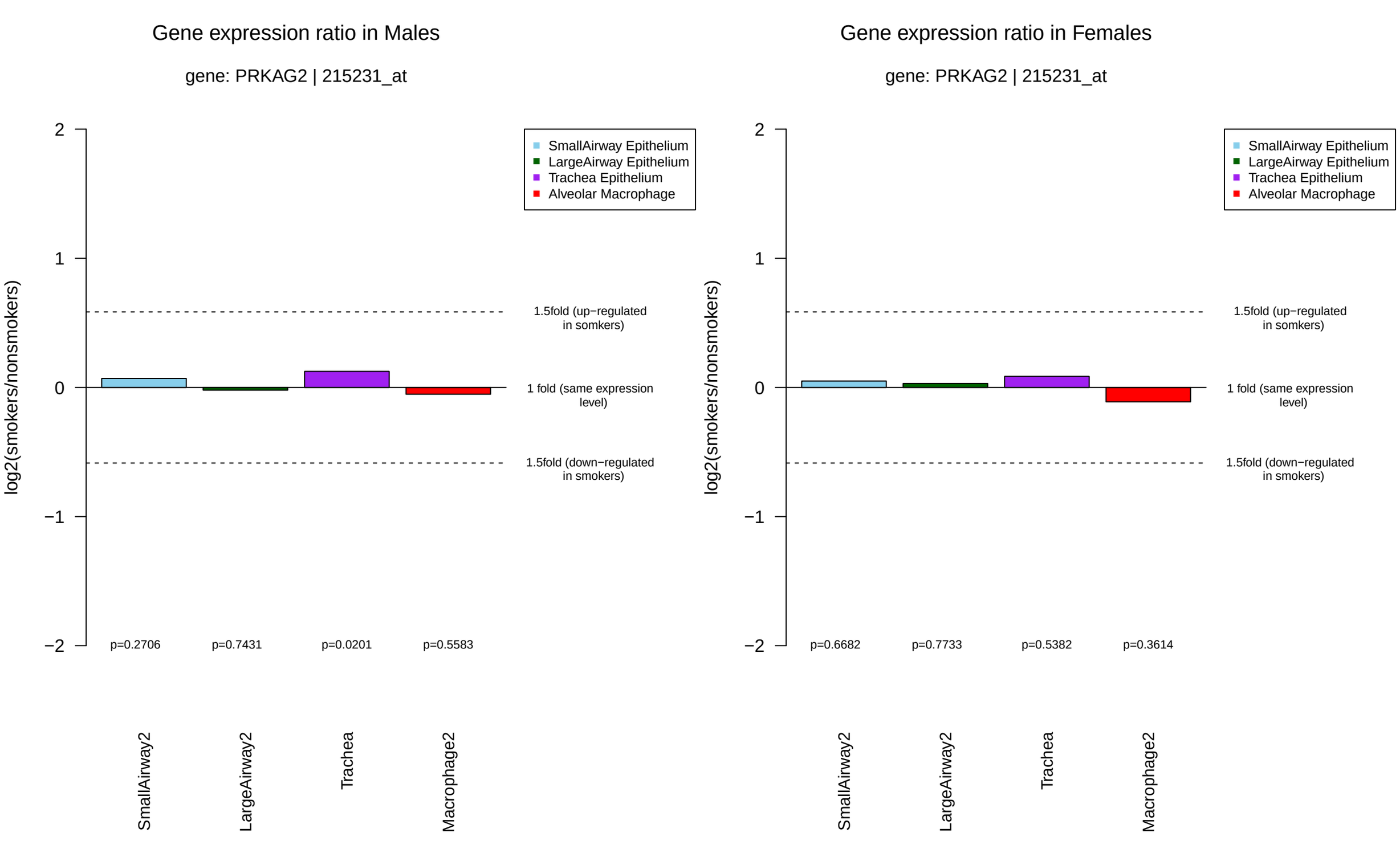
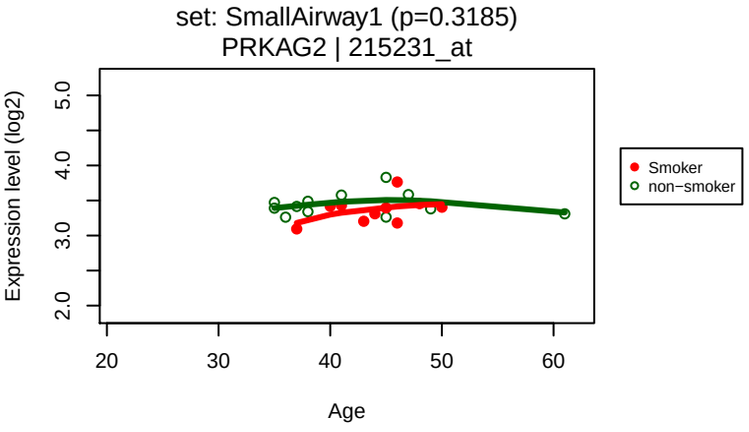
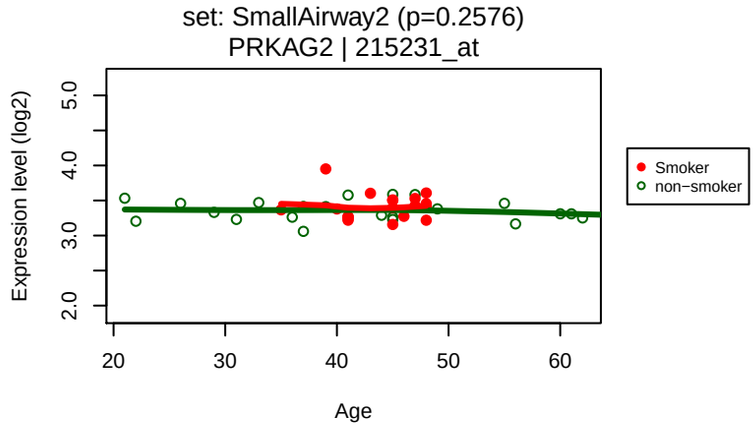
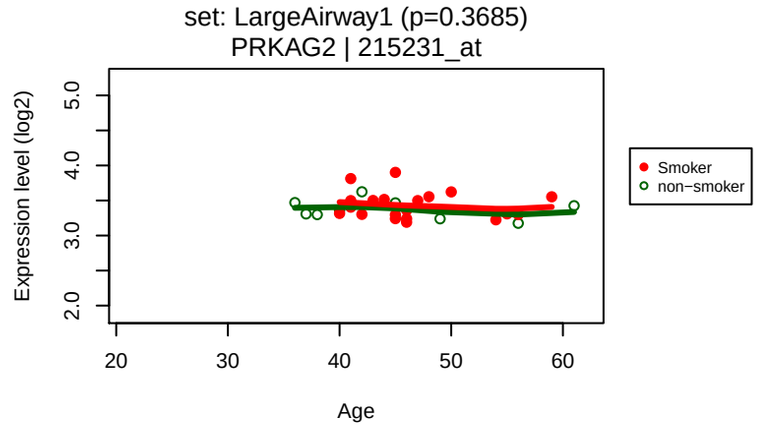

Gene expression of smokers and nonsmokers
gene: PRKAG2 | 215231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3185
p=0.2576
p=0.1454
p=0.3685
p=0.7952
p=0.0127
p=0.7969
p=0.2908
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

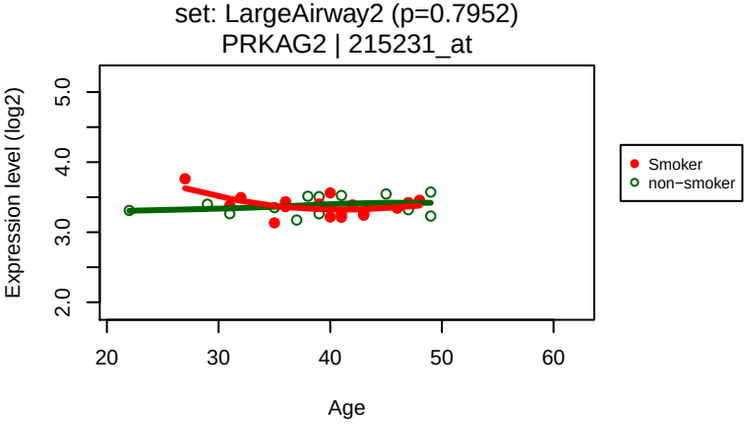

Gene expression ratio (smokers/nonsmokers)
gene: PRKAG2 | 215231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3185
p=0.2576
p=0.1454
p=0.3685
p=0.7952
p=0.0127
p=0.7969
p=0.2908
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

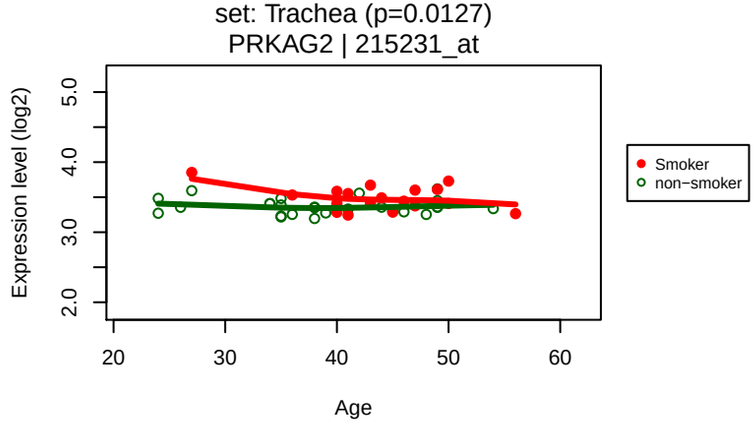

Gene expression ratio in Males
gene: PRKAG2 | 215231_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2706
p=0.7431
p=0.0201
p=0.5583
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAG2 | 215231_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6682
p=0.7733
p=0.5382
p=0.3614
SmallAirway2
LargeAirway2
Trachea
Macrophage2

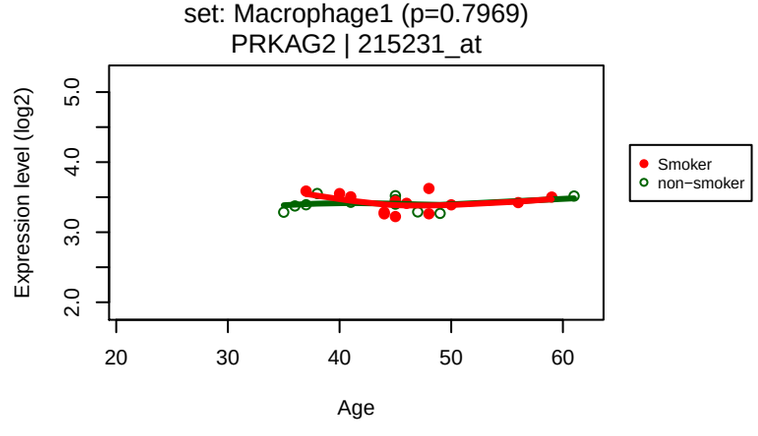

set: SmallAirway1 (p=0.3185)
PRKAG2 | 215231_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

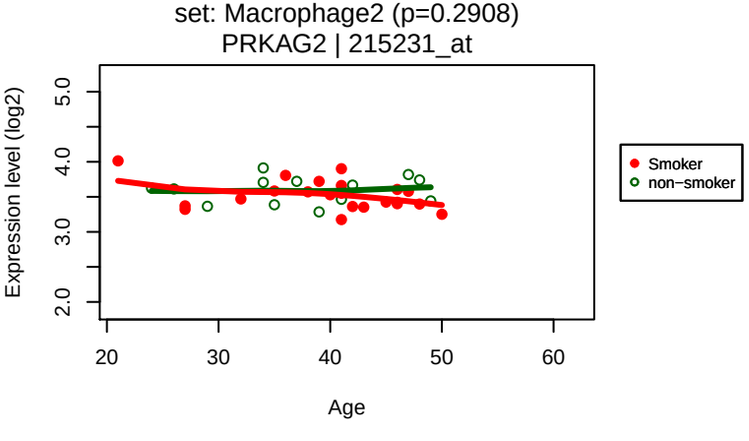

set: SmallAirway2 (p=0.2576)
PRKAG2 | 215231_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

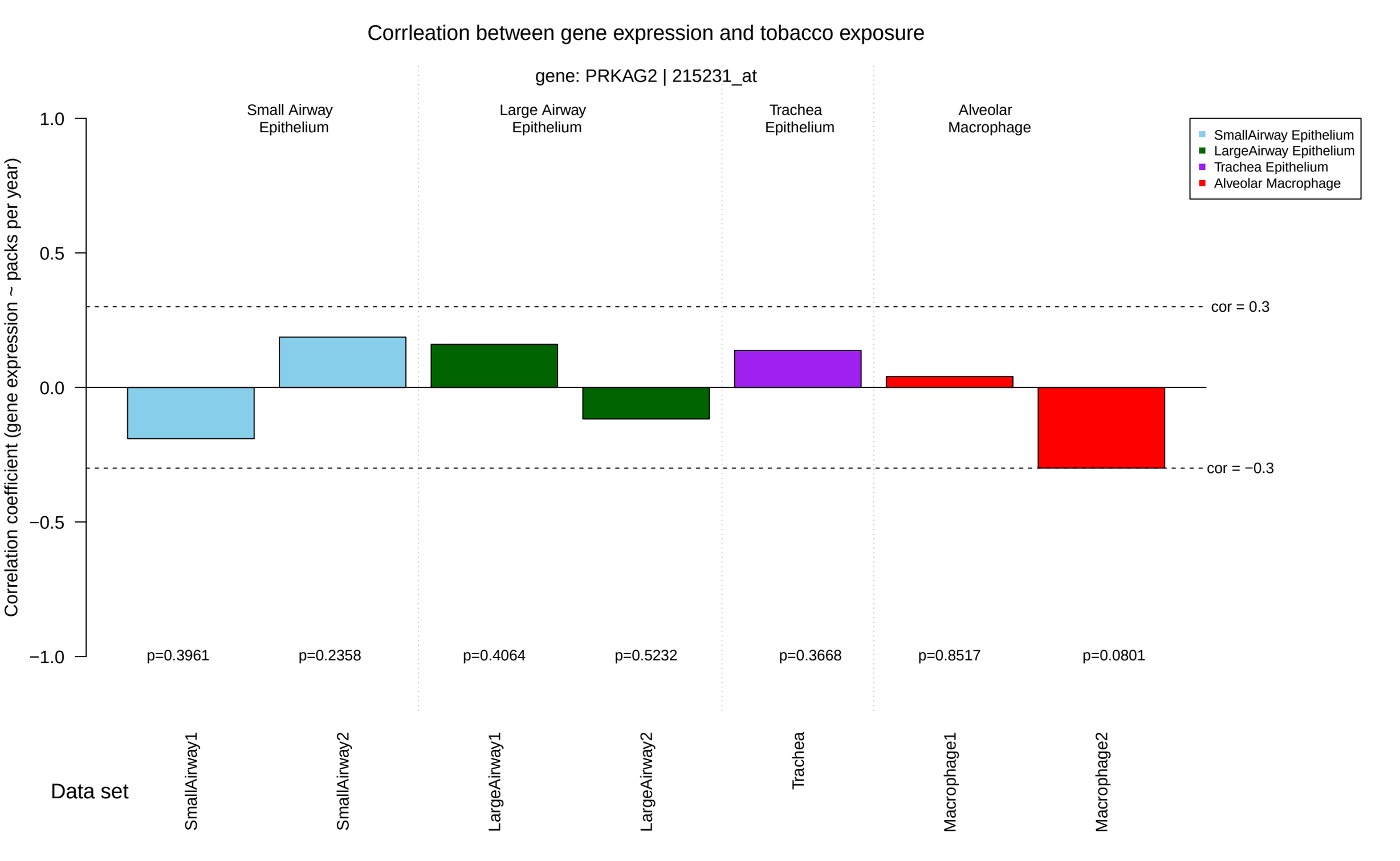

set: LargeAirway1 (p=0.3685)
PRKAG2 | 215231_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7952)
PRKAG2 | 215231_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0127)
PRKAG2 | 215231_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7969)
PRKAG2 | 215231_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2908)
PRKAG2 | 215231_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAG2 | 215231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3961
p=0.2358
p=0.4064
p=0.5232
p=0.3668
p=0.8517
p=0.0801
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2